

llmXive Automated Review of Weak-to-Strong Generalization via Direct On-Policy Distillation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a novel method (Direct-OPD) for weak-to-strong generalization, and the core experimental claims regarding the performance gains of the proposed method (e.g., Qwen3-1.7B improving from 48.3% to 58.3% on AIME 2024) are well-supported by the internal data in Table 1 and the corresponding figures. The logical derivation of the policy-shift-as-reward mechanism is sound and consistent with the cited literature on KL-regularized RL.

However, there are specific instances where quantitative claims regarding external baselines and resource costs are stated with high confidence but lack direct evidentiary support within the provided text or tables. Specifically, the comparison to the “Polaris” baseline (Abstract, Intro) and the precise wall-clock time estimates for the RL baselines (Section 4.2) are presented as facts without a corresponding data point in the results or a verifiable citation for the external resource usage. While the relative efficiency argument is strong, the specific numbers (e.g., “1 week,” “160 hours”) are currently unsupported load-bearing claims.

Additionally, the bibliography includes references to models with future dates (DeepSeek V4, GLM-5.2, both 2026). Unless these are explicitly marked as hypothetical or internal previews with a verifiable source, citing non-existent future models as established “frontier systems” constitutes a citation accuracy issue that undermines the credibility of the related work survey. These items require correction to ensure all factual claims are strictly supported by the provided evidence or verifiable external sources.

Action items.

- **[writing]** The paper presents a novel method (Direct-OPD) for weak-to-strong generalization, and the core experimental claims regarding the performance gains of the proposed method (e.g., Qwen3-1.7B improving from 48.3% to 58.3% on AIME 2024) are well-supported by the internal data in Table 1 and the corresponding figures. The logical derivation of the policy-shift-as-reward mechanism is sound and consistent with the cited literature on KL-regularized RL. However, there are specific instances where quantitat

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure fails to match its caption, which describes two separate panels for different teacher pairs, whereas the image shows a single combined plot. Additionally, the legend labels and axis text contradict the specific model pairs and metrics defined in the caption.

Figure 2

The figure effectively visualizes the entropy diagnostics described in the caption, showing the expected trends of non-collapsing actor entropy and narrowing teacher/reference gaps. However, the top-left subplot title is redundant rather than descriptive, and the y-axis tick labels on the top row are rendered too small for comfortable reading.

Figure 3

The figure effectively displays the response-length sweep results with clear legends and gridlines, but the y-axis labels are rotated and rendered in a font size that is difficult to read.

Figure 4

The figure is visually clear with a legend, but the provided caption is truncated and contains a typo in the mathematical definition. Additionally, the legend includes a ‘6k’ condition that appears to behave as a static baseline in the middle panel without explanation.

Figure 5

The figure effectively demonstrates the impact of KL coefficients and adaptive KL on validation accuracy and token reward. However, the top row y-axis label is ambiguous regarding the ‘AIME 2024/2025’ metric, and the split legends across subplots are redundant.

Figure 6

The figure presents four entropy diagnostic plots with clear titles and axes, but the caption contains a broken cross-reference to a missing figure number, and the plot titles lack explicit unit labels.

Figure 7

The figure presents transfer curves for the QuestA setting on AIME 2025, but the legend uses inconsistent abbreviations (‘R1-7B’ vs ‘R1-Distill-7B’) and fails to explicitly state the numerical value of the ‘Initial’ baseline.

Figure 8

The figure effectively displays transfer learning results across different models, but the caption contains significant grammatical errors and missing words. Additionally, the legend definition for ‘Initial Student’ is ambiguous given the presence of multiple baseline lines in the plots.

Figure 9

Figure 9 presents a compelling comparison of training efficiency, but the caption is truncated and the legend for the rightmost subplot is incomplete, leaving the meaning of the ‘0.635’ baseline

undefined.

Figure 10

The figure is partially rendered, missing the ‘Right’ panel of endpoint scores described in the caption. Additionally, specific labels like the ‘Switch to QuestA’ annotation and the figure title are not defined in the caption text.

Figure 11

Figure 11 effectively illustrates the performance gains of the proposed method over vanilla OPD, but the legend in panel (a) mislabels the proposed method as ‘Direct-OPD’ (contradicting the caption), and panel (b) lacks clear y-axis tick labels for precise value reading.

Action items.

- **[science]** Figure 1: The caption claims to show two panels (‘Left: R1-Distill...’ and ‘Right: Nemotron...’), but the rendered image displays only a single plot titled ‘JustRL overlap dynamics’ containing both transfer pairs. The figure fails to visually separate the two distinct teacher-student pairs as described.
- **[writing]** Figure 1: The legend labels ‘JustRL -> R1-7B’ and ‘JustRL -> Qwen3-1.7B’ do not match the caption’s description of the ‘Left’ pair (‘R1-Distill-1.5B -> JustRL-1.5B’). The axis label ‘Student top-k overlap ratio’ is also inconsistent with the caption’s ‘Teacher-student top-k overlap’.
- **[science]** Figure 2: The top-left subplot is titled ‘Qwen3-1.7B’ but the caption identifies the first column as ‘student entropy’. Since the student is Qwen3-1.7B, the title is redundant and potentially confusing; it should explicitly label the metric (e.g., ‘Student Entropy’) to match the caption’s description of the columns.
- **[writing]** Figure 2: The y-axis tick labels on the top row (0.225, 0.300, 0.375, 0.450) are rendered in a font size that is significantly smaller and less legible than the x-axis labels or subplot titles, reducing readability.
- **[writing]** Figure 3: The y-axis label ‘AIME 2024/2025’ is illegible due to vertical rotation and small font size, making it difficult to read the metric name.
- **[writing]** Figure 3: The y-axis label ‘avg. accuracy (ave@32)’ is rotated 90 degrees and compressed, reducing readability.
- **[fatal]** Figure 4: The provided caption is truncated mid-sentence (‘...actors tra’), failing to describe the middle and right panels shown in the image.
- **[science]** Figure 4: The legend defines ‘6k fixed KL1’ (pink), but the middle panel’s pink curve remains flat at 0 while the caption implies all actors should show drift; the legend does not explain this baseline behavior.
- **[writing]** Figure 4: The mathematical definition of the cumulative gap G_T in the caption contains a typo (‘=_t T g_t’) and is missing the summation symbol and index range.

- **[science]** Figure 5: The top row y-axis label ‘AIME 2024/2025’ contradicts the caption’s claim that the plot shows ‘AIME 2024/2025 validation accuracy (ave@32)’; the axis label implies a single metric combining both years, while the caption suggests the plot represents the average of the two. This ambiguity makes it unclear if the data is an aggregate or a specific year.
- **[writing]** Figure 5: The legend in the top row is split across three separate panels with inconsistent entries (e.g., ‘KL 0.8’ only in the right panel), which is redundant and cluttered. A single unified legend or consistent labeling across all subplots would improve readability.
- **[writing]** Figure 6: The caption contains a broken cross-reference (‘Together with Figure , this shows...’) where the figure number is missing.
- **[science]** Figure 6: The plot titles (‘Qwen3-1.7B’, ‘QuestA teacher’, ‘QuestA teacher ref’, ‘QuestA delta’) do not explicitly label the y-axis units (entropy bits) or confirm the metric is entropy, relying entirely on the caption for context.
- **[science]** Figure 7: The legend defines ‘QuestA -> R1-7B’ and ‘QuestA -> Qwen3-1.7B’, but the plot titles are ‘R1-Distill-7B’ and ‘Qwen3-1.7B’. The legend label ‘R1-7B’ is an inconsistent abbreviation that does not match the full model name in the title.
- **[science]** Figure 7: The legend entry ‘Initial’ corresponds to a dashed line, but the legend does not specify the numerical value of this baseline, forcing the reader to estimate it from the y-axis.
- **[writing]** Figure 8: The caption contains multiple grammatical fragments and missing terms (e.g., ‘transfers RL-induced...’, ‘showing that is not specific...’), making the text difficult to read.
- **[science]** Figure 8: The legend defines ‘Initial Student’ as a dashed line, but the plots contain multiple horizontal lines (dashed, dotted, and solid) representing baselines (e.g., ‘JustRL 0.512’), creating ambiguity about which line corresponds to the legend entry.
- **[writing]** Figure 8: The y-axis label ‘AIME 2024 Accuracy (ave@32)’ is repeated for the top row, but the bottom row y-axis label is ‘AIME 2025 Accuracy (ave@32)’; however, the caption text ‘evaluated on AIME 2024 and AIME 2025’ is vague about which specific subplot corresponds to which year without explicit row headers.
- **[science]** Figure 9: The legend defines ‘T1500’ (orange line) but the caption text is truncated (‘...short transf’), failing to define what T1500 represents or how it relates to the ‘TN’ notation described.
- **[science]** Figure 9: The rightmost subplot (‘Qwen3 nonthinking: AIME 2025’) contains a dashed blue line labeled ‘0.635’ but lacks a corresponding legend entry or caption definition explaining what this baseline represents.
- **[writing]** Figure 9: The caption text is truncated at the end (‘...short transf’), cutting off the description of the wall-clock time calculation for the transfer points.

- **[science]** Figure 10: The caption states the figure contains ‘Left: trajectories... Right: endpoint scores’, but the rendered image displays only a single plot of trajectories (steps 0–600) and lacks the promised ‘Right’ panel showing endpoint scores.
- **[writing]** Figure 10: The title ‘Qwen3-1.7B: JustRL then QuestA policy-shift transfer’ and the vertical dashed line at step 300 labeled ‘Switch to QuestA’ are not explicitly defined in the figure’s own caption, which only describes the signals generally.
- **[science]** Figure 11(a): The caption claims the blue curve ‘transfers the JustRL-1.5B - R1-Distill-1.5B policy shift,’ but the legend labels it ‘Direct-OPD.’ This contradicts the paper’s distinction between ‘vanilla OPD’ (red) and the proposed method, creating ambiguity about whether the blue curve represents the proposed method or a different baseline.
- **[writing]** Figure 11(a): The y-axis label ‘AIME 24 ave@32’ is truncated and difficult to read due to tight spacing; expand the label or increase font size for clarity.
- **[writing]** Figure 11(b): The bar chart lacks explicit y-axis tick labels (only gridlines are visible), making it hard to verify the exact values of the bars and the ‘Teacher’ reference line.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper demonstrates excellent adherence to the “adjacent-field PhD” standard regarding terminology and notation. The authors are rigorous in defining their core contributions and specific symbols at their first occurrence.

Specifically, the central concept of “Direct On-Policy Distillation” and its acronym `\method{}` (Direct-OPD) are explicitly defined in the Abstract and Introduction. The mathematical notation for the policy shift (Δ_T , π_T , $\pi_{T_{\text{ref}}}$) is introduced with clear prose definitions in the Introduction (Eq. 1) and formalized in Section 2 (Eq. 2), ensuring a reader from a neighboring field (e.g., standard RL or NLP) can follow the derivation without guessing.

The paper also handles subfield-specific acronyms well. Terms like “RLVR” (Reinforcement Learning with Verifiable Rewards) and “OPD” (On-Policy Distillation) are either spelled out immediately upon first use or are standard enough within the broader ML community that their usage does not constitute a barrier. The distinction between the “teacher” and “student” policies, and the specific “top- k ” approximation used, is explained operationally rather than assumed.

There are no instances of undefined symbols, overloaded notation, or “lab slang” that would cause a competent reader to stall. The use of “policy shift” is consistently defined as the log-ratio of the post-RL and pre-RL policies, preventing ambiguity. The paper is self-contained regarding its specific vocabulary.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper’s argument structure is logically sound and internally consistent. The central thesis—that the policy shift ($\log \pi_T - \log \pi_{T_{\text{ref}}}$) serves as a transferable implicit reward distinct from the teacher’s final policy—is defined clearly in Section 2 (Eq. 1) and consistently applied throughout the methodology and experiments.

The logical flow from premises to conclusions holds:

1. **Premise:** Standard OPD fails in weak-to-strong settings because it imitates the teacher’s capacity-limited final policy (Section 1, Fig 1a).
2. **Mechanism:** The proposed method isolates the RL-induced shift, which mathematically corresponds to the implicit reward under KL-regularized RL (Section 2.2, Eq. 2-3).
3. **Evidence:** Experiments show that applying this shift improves students stronger than the teacher (Section 4.1, Table 1a), whereas standard OPD degrades them. This directly supports the claim that the *shift* is the transferable object, not the *policy*.
4. **Consistency:** The analysis in Section 5 confirms the mechanism by showing that transfer occurs even without high token-overlap (Fig 2), validating the “shift” hypothesis over the “imitation” hypothesis.

There are no contradictions between sections. The limitations section (Section 6) appropriately qualifies the claims made in the conclusion, noting that the signal is conditional on the meaningfulness of the teacher’s shift, which aligns with the analysis in Section 5.3 regarding KL control and reward reliability. The numerical claims in the abstract (48.3% to 58.3%) match the results in Table 1a and the introduction exactly. The definitions of π_T , $\pi_{T_{\text{ref}}}$, and the student π_θ remain stable across the text. No non-sequiturs or scope inflation were detected.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper presents a compelling method for weak-to-strong generalization, but the rhetoric in the abstract, introduction, and conclusion occasionally exceeds the specific scope of the experimental evidence provided.

The primary issue lies in the generalization of specific efficiency and performance gains. The abstract highlights a specific result: “boosts Qwen3-1.7B from 48.3% to 58.3% on AIME 2024 in

just 4 hours.” While this result is accurate for the JustRL teacher pair (Table 1a), the phrasing presents it as a headline capability of the method without immediately qualifying that this specific efficiency profile is tied to the specific teacher-student configuration tested. The method’s performance and speed are shown to be sensitive to the teacher pair and KL coefficient (Section 5.4), so presenting a single data point as the definitive “efficiency” of the method is slightly overreaching.

Similarly, the Introduction compares the method’s efficiency to “Polaris” running for “at least a week.” This is a comparison to an external report rather than a controlled, internal experiment where the authors ran direct RL on the same hardware for the same duration. While the direction of the argument is sound, framing it as a direct “comparable” result without the authors’ own matched baseline on the same infrastructure risks overstating the empirical basis of the efficiency claim.

Finally, the Conclusion states the method “outperforms step-matched direct RL.” The evidence for this (Section 4.2) is limited to two specific transfer scenarios (R1-Distill 1.5B->7B and Qwen3 1.7B->4B). The claim is presented as a general property of the method, whereas the data only supports it for the specific model families and scales tested. The paper would be more rigorous by explicitly limiting this claim to the tested regimes or acknowledging that the superiority is demonstrated within the specific experimental bounds.

These are primarily issues of scope and framing rather than factual errors. The results are strong, but the language should be tightened to ensure the claims track precisely with the boundaries of the experiments performed.

Action items.

- **[writing]** Abstract: The claim ‘boosts Qwen3-1.7B... in just 4 hours’ presents a specific result from one teacher pair as a general method capability. Scope this to ‘in our primary setting’ or clarify it applies to the JustRL pair to avoid implying universal efficiency.
- **[writing]** Introduction: The comparison to Polaris (‘at least a week’) relies on external estimates rather than an internal matched experiment. Clarify this is a comparison to reported external results, not a direct internal ablation, to avoid overstating the empirical basis.
- **[writing]** Conclusion: The claim ‘outperforms step-matched direct RL’ is broad. The data only supports this for specific pairs (R1-Distill 1.5B->7B, Qwen3 1.7B->4B). Narrow the claim to ‘in our tested teacher-student pairs’ to match the evidence scope.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This paper presents a method for weak-to-strong generalization in mathematical reasoning, utilizing a “policy shift” derived from a smaller model’s reinforcement learning (RL) run to

guide a larger model. The research focuses on algorithmic efficiency and performance gains on standard benchmarks (AIME 2024/2025).

From a safety and ethics perspective, the work is low-risk. The methodology does not involve:

1. **Human Subjects:** The data used (Skywork-OR1-RL-Data, DAPO-Math-17K) consists of mathematical problems and solutions, not personal data, private communications, or human behavioral logs requiring IRB approval.
2. **Dual-Use for Harm:** The method optimizes for mathematical reasoning capabilities. It does not lower the barrier for generating disinformation, cyberattacks, biological synthesis, or other harmful content. The “policy shift” is a mathematical abstraction of RL updates, not a mechanism for deception or manipulation.
3. **Data Privacy/PII:** No Personally Identifiable Information (PII) is released or discussed. The datasets are standard public benchmarks for math reasoning.
4. **Vulnerable Populations:** The study does not involve children, patients, or other protected groups.

The paper includes a “Limitations” section (Section 6) acknowledging technical constraints (e.g., signal reliability, hyperparameter sensitivity) but does not require an expanded ethical disclosure because the research topic (math reasoning optimization) does not inherently carry foreseeable, non-trivial risks of harm that are currently unmitigated. The use of public, pre-existing datasets and the focus on a benign capability (math) align with standard low-risk ML research norms. No specific safety disclosures or mitigations are missing.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling method for weak-to-strong generalization, but the experimental design has gaps that prevent the reported results from definitively establishing the claimed mechanisms.

First, the primary evidence for the method’s efficacy relies on single-run results reported in Table 1 and Figure 2. For instance, the claim that Direct-OPD boosts Qwen3-1.7B from 48.3% to 58.3% on AIME 2024 is presented as a definitive improvement. However, Reinforcement Learning with Verifiable Rewards (RLVR) is notoriously stochastic; performance can vary significantly across different random seeds due to the high variance in rollout generation and reward signals. Without reporting results across multiple seeds (e.g., 3-5) with standard deviations or confidence intervals, it is impossible to distinguish a genuine methodological improvement from a lucky seed or sampling noise. A 10% point gain is substantial, but if the standard deviation across seeds is 5-8%, the result is not statistically robust. The authors must report variance to validate

that the effect is reproducible.

Second, the efficiency claim in Section 4.2 (“Weak-to-strong generalization beats RL”) compares wall-clock time but fails to control for total compute or data exposure. The authors compare a 1.5B model trained for 1500 steps (plus a short transfer) against a 7B model trained for a matched number of steps. However, the 1.5B model likely consumes significantly fewer FLOPs per step. If the 7B model were allowed to run for a compute budget equivalent to the 1.5B model’s training plus the transfer cost, it might achieve similar or better performance. The current design confounds the “weak-to-strong” mechanism with the possibility that the small model simply had a more favorable training trajectory or data exposure per unit of compute. A compute-matched ablation is required to isolate the benefit of the transfer mechanism itself.

Finally, the sequential composition experiment (Section 4.3) lacks a negative control. The authors show that applying Shift A then Shift B improves performance. However, they do not demonstrate that applying Shift A then Shift A (or a random shift) does not yield similar gains. Without this control, the improvement could be attributed to the simple fact of additional training steps (more optimization) rather than the specific composition of distinct policy shifts. A control run applying the same shift twice would help rule out the “more training” alternative explanation.

Action items.

- **[writing]** The paper presents a compelling method for weak-to-strong generalization, but the experimental design has gaps that prevent the reported results from definitively establishing the claimed mechanisms. First, the primary evidence for the method’s efficacy relies on single-run results reported in Table 1 and Figure 2. For instance, the claim that Direct-OPD boosts Qwen3-1.7B from 48.3% to 58.3% on AIME 2024 is presented as a definitive improvement. However, Reinforcement Learning with Verifiable Re

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The paper presents a novel method for weak-to-strong generalization, but the statistical reporting of quantitative results is insufficient to support the strength of the claims.

Missing Uncertainty Reporting: Throughout Section 4 and Table 1, the authors report single-point accuracy metrics (e.g., “58.3% on AIME 2024”) without mentioning the number of random seeds used or providing standard deviations (SD), standard errors (SE), or confidence intervals (CI). In LLM training, performance varies due to initialization and sampling stochasticity. Reporting a single number implies unjustified precision. The field standard is to report mean $\pm$ SD over at least 3 independent seeds. Without this, it is impossible to determine if improvements (e.g., +10.0%) are robust or artifacts of a specific seed.

Comparative Claims Without Variance: In Section 4.2, the claim that Direct-OPD

“outperforms step-matched direct RL” relies on comparing single trajectories in Figure 3. Since both methods are stochastic, a single-run comparison is statistically weak. To validly claim one method “outperforms” another, the authors must demonstrate consistent performance differences across multiple runs. The current presentation precludes hypothesis testing (e.g., paired t-tests) and does not allow readers to assess distribution overlap.

Recommendation: The authors should re-run main experiments with at least 3 random seeds and report mean $\pm$ SD in all tables and figures. If variance is low, point estimates may suffice with a note; if high, the “outperforms” claim requires statistical support. This is a writing fix if data exists, or a science fix if new runs are needed.

Action items.

- **[writing]** Table 1 and Section 4.1 report single-point accuracy (e.g., 58.3%) without uncertainty metrics (SD/SE/CI) or seed counts. Report mean $\pm$ SD over 3 seeds for all main results to distinguish signal from noise.
- **[science]** Section 4.2 claims Direct-OPD ‘outperforms’ direct RL based on single runs. Report mean $\pm$ SD over 3 seeds for both methods to statistically validate the comparative claim.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper is generally well-written, with a clear logical flow and precise technical exposition. The abstract effectively summarizes the method and key results. However, there are a few instances where sentence structure impedes immediate comprehension, requiring the reader to re-parse or untangle clauses.

In Section 4.1, the second paragraph contains a run-on sentence that attempts to define the second experimental setup and its limitations simultaneously. The clause “This setting is not intended as a strict weaker-teacher-to-stronger-student comparison...” is buried after a comma, making the sentence’s primary purpose (describing the test) less distinct from its secondary purpose (defining the scope). Splitting this into two sentences would clarify the experimental design.

Similarly, in Section 5.2, the second paragraph features a comma splice joining two independent clauses regarding the efficiency of small models and the transfer mechanism. This creates a slight rhythmic stumble. A simple punctuation change or restructuring into a relative clause would resolve this.

In Section 5.3, the description of the “natural concern” regarding short-horizon training is followed by a very long sentence explaining the test methodology. While grammatically correct, the density of the second sentence (“We test this by evaluating...”) makes it slightly harder to parse on the first pass. Breaking this into two shorter sentences would improve readability.

without losing technical precision.

Finally, in Section 5.4, the explanation of KL coefficient sensitivity contains a slightly ambiguous phrase regarding probability assignment. The phrase “where both teacher and reference assign low probability” could be misread as a separate clause rather than a modifier for “regions.” A minor rephrasing to explicitly link the probability assignment to the regions would eliminate this potential confusion.

These are minor issues that do not obscure the scientific contribution but, if addressed, would allow the reader to move through the text with even greater fluidity.

Action items.

- **[writing]** The paper is generally well-written, with a clear logical flow and precise technical exposition. The abstract effectively summarizes the method and key results. However, there are a few instances where sentence structure impedes immediate comprehension, requiring the reader to re-parse or untangle clauses. In Section 4.1, the second paragraph contains a run-on sentence that attempts to define the second experimental setup and its limitations simultaneously. The clause “This setting is not intend